

Guangxu Yang (杨广绪)

PhD Student in Computer Science

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Location: Los Angeles, USA

Education

University of Southern California (USC)

Ph.D. Student in Computer Science

Advisor: Prof. Jiapeng Zhang

Los Angeles, USA

Aug 2022 – Dec 2026

University of Electronic Science and Technology of China (UESTC)

M.E. in Electrical Engineering

B.E. in Electrical Engineering

Chengdu, China

Sep 2019 – Jun 2022

Sep 2015 – Jun 2019

Research Interests

- Provable and practical quantum advantages
- Communication complexity

Publications & Preprints

Note: Authors in theoretical computer science are listed alphabetically.

- [1] **Exponential Quantum Advantages for Decoded Quantum Interferometry in the Streaming Setting**
Kewen Wu, **Guangxu Yang**. [arXiv:2610.01902](#)
- [2] **Exponential Quantum Space Advantage for Approximating Max- k -SAT in the Streaming Setting**
Haoyu Wang, **Guangxu Yang**. [arXiv:2606.05366](#)
- [3] **Exponential Quantum Advantage in Numbers-on-Forehead Communication**
Haoyu Wang, Pei Wu, **Guangxu Yang**. [arXiv:2609.40273](#)
- [4] **Quantum Communication Lower Bounds for Search Problems via Matrix Discrepancy**
Minbo Gao, Chenghua Liu, **Guangxu Yang**, Tianyi Zhang. [arXiv:2607.08517](#)
38th ACM-SIAM Symposium on Discrete Algorithms (SODA 2027).
- [5] **Prediction Under Imperfect Compression: A Theory of Approximate MDL**
Qian Li, Xinyu Mao, Shanghua Teng, **Guangxu Yang**. [arXiv:2606.04834](#)
The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026 oral, top 1.4% of accepted papers).
- [6] **Exponential Separation of Quantum and Classical One-Way Numbers-on-Forehead Communication**
Guangxu Yang, Jiapeng Zhang. [arXiv:2603.22795](#)
67th Annual Symposium on Foundations of Computer Science (FOCS 2026).
- [7] **A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity**
Xudong Wu, **Guangxu Yang**, Penghui Yao. [arXiv:2511.17227](#)
53rd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2026).
- [8] **Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication**
Guangxu Yang, Jiapeng Zhang. [arXiv:2506.12420](#).

- [9] **A Min-Entropy Approach to Multi-Party Communication Lower Bounds**
Mi-Ying Huang, Xinyu Mao, Shuo Wang, [Guangxu Yang](#), Jiapeng Zhang.
40th Computational Complexity Conference (CCC 2025).
- [10] **Gadgetless Lifting Beats Round Elimination: Improved Lower Bounds for Pointer Chasing**
Xinyu Mao, [Guangxu Yang](#), Jiapeng Zhang. [arXiv:2411.10996](#)
16th Innovations in Theoretical Computer Science (ITCS 2025).
- [11] **Communication Lower Bounds for Collision Problems via Density Increment Arguments**
[Guangxu Yang](#) and Jiapeng Zhang.
56th Annual ACM Symposium on Theory of Computing (STOC 2024).
- [12] **Communication Lower Bounds of Key-Agreement Protocols via Density Increment Arguments**
Mi-Ying Huang, Xinyu Mao, [Guangxu Yang](#), Jiapeng Zhang. [eprint:2023.1349](#)
21st Theory of Cryptography Conference (TCC 2023).

Teaching Experience

Teaching Assistant at USC

- **CSCI 670:** Advanced Analysis of Algorithms *Fall 2026*
- **CSCI 570:** Analysis of Algorithms *Spring 2026*
- **CSCI 170:** Discrete Methods in Computer Science *Spring 2025*
- **CSCI 575:** Quantum Computing and Quantum Cryptography *Spring 2024*

Professional Service

Conference Reviewer: STOC 2027, ITCS 2027, SOSA 2027, SODA 2027, MFCS 2026, FOCS 2026, PODS 2026, CRYPTO 2025, ITCS 2025, COCOON 2024.

Presentations

Conference Talks

Communication Lower Bounds of Key-Agreement Protocols

TCC 2023, Taipei Nov 2023

Gadgetless Lifting Beats Round Elimination

ITCS 2025, New York Jan 2025

Invited Seminar & Group Talks

Communication Lower Bounds for Collision Problems via Density Increment Arguments

University of Southern California Theory Lunch	Oct 2023
Nanjing University	Dec 2023
Shanghai Jiao Tong University	May 2024
Shanghai University of Finance and Economics	Oct 2025

Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication

University of Southern California Theory Lunch	May 2025
University of Science and Technology of China	Nov 2025
SoCal Theory Day at University of California San Diego	Feb 2026
Shanghai Jiao Tong University	Mar 2026

Complete List of Publications

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- [1] **Exponential Quantum Advantages for Decoded Quantum Interferometry in the Streaming Setting**
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40th Computational Complexity Conference (CCC 2025).
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21st Theory of Cryptography Conference (TCC 2023).

Research Plan

My research studies provable and practical quantum advantages under communication and memory constraints. I combine quantum algorithm design with complexity-theoretic lower bounds to identify problems for which quantum resources offer substantial benefits. Communication and streaming models allow exponential advantages to be proved unconditionally. Building on this foundation, I plan to pursue two complementary directions: establishing conditional quantum speedups for natural optimization problems and developing quantum streaming algorithms with practical resource requirements.

Research foundation

With collaborators, I have established exponential quantum advantages in one-way and general interactive three-party numbers-on-forehead communication [3, 6], and developed quantum streaming algorithms for approximating Max- k SAT [2]. In recent work with Kewen Wu, we adapted decoded quantum interferometry (DQI) to the streaming setting, obtaining an exponential space advantage for a one-pass quantum algorithm over multi-pass classical algorithms [1]. My work on gadgetless lifting and density-increment methods provides a foundation for proving communication lower bounds [10, 11], while complementary work on matrix discrepancy develops techniques for quantum lower bounds [4].

1. From space advantages to conditional quantum speedups

DQI connects approximate optimization to classical error correction. For Optimal Polynomial Intersection (OPI), it provides approximation guarantees beyond those achieved by known polynomial-time classical algorithms in certain parameter regimes. I aim to identify OPI-related optimization problems that admit polynomial-time quantum algorithms based on DQI while remaining hard for classical polynomial-time algorithms under explicit complexity-theoretic or cryptographic assumptions.

Beyond analyzing existing algorithms, I also aim to develop quantum algorithms for other natural problems. I will combine classical hardness analysis with concrete resource estimates to identify settings in which exponential time advantages could translate into practical benefits. This direction extends my study of unconditional space advantages toward time advantages with clearly stated assumptions.

2. Toward practical quantum streaming algorithms

I plan to develop implementation-oriented variants of our Max- k SAT algorithm and related Max-DiCut algorithms. A proof-of-principle implementation of the quantum pair sketch for Hidden Matching on a trapped-ion processor provides a starting point (Niroula et al., [arXiv:2511.03689](#)). Its accompanying fault-tolerant resource estimates identify circuit depth as a major obstacle to practical advantage.

I will focus on reducing logical-qubit requirements and circuit depth, improving noise robustness, and minimizing latency from classical control. The aim is to preserve useful approximation guarantees while making the required quantum computation more feasible. Concrete resource estimates will guide which algorithmic improvements have the greatest practical effect. By connecting implementation considerations with rigorous algorithm analysis and classical lower bounds, I seek to translate asymptotic space savings into feasible quantum streaming algorithms.